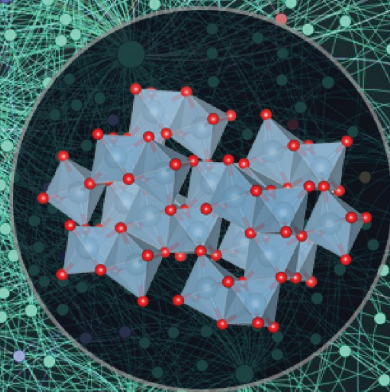
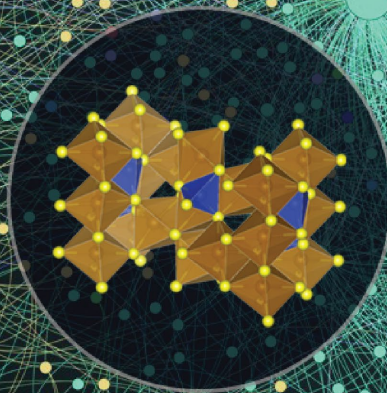


splitting data into test, train, validation



We need a test set to prevent p-hacking!

What is p-hacking?

<u>P-VALUE</u>	<u>INTERPRETATION</u>
0.001	HIGHLY SIGNIFICANT
0.01	
0.02	
0.03	
0.04	SIGNIFICANT
0.049	
0.050	OH CRAP. REDO CALCULATIONS.
0.051	ON THE EDGE OF SIGNIFICANCE
0.06	
0.07	HIGHLY SUGGESTIVE, SIGNIFICANT AT THE $P < 0.10$ LEVEL
0.08	
0.09	
0.099	HEY, LOOK AT THIS INTERESTING SUBGROUP ANALYSIS
≥ 0.1	

https://imgs.xkcd.com/comics/p_values.png



Recall our linear models

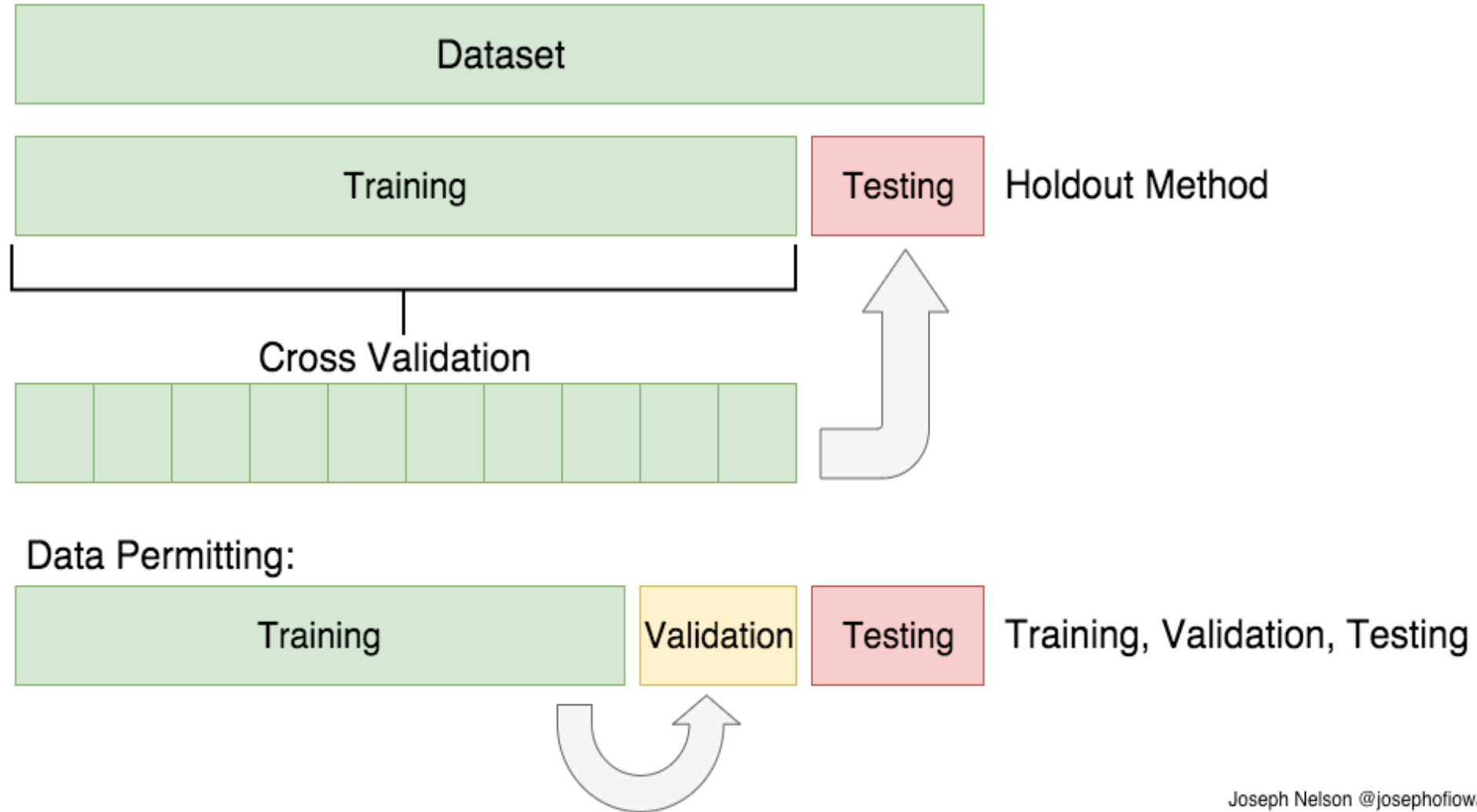
$$\textit{dependent variable} = \textit{constant} + \textit{parameter} * IV + \textit{parameter} * IV + \dots$$

If a constant is zero, then that parameter doesn't contribute to the dependent variable

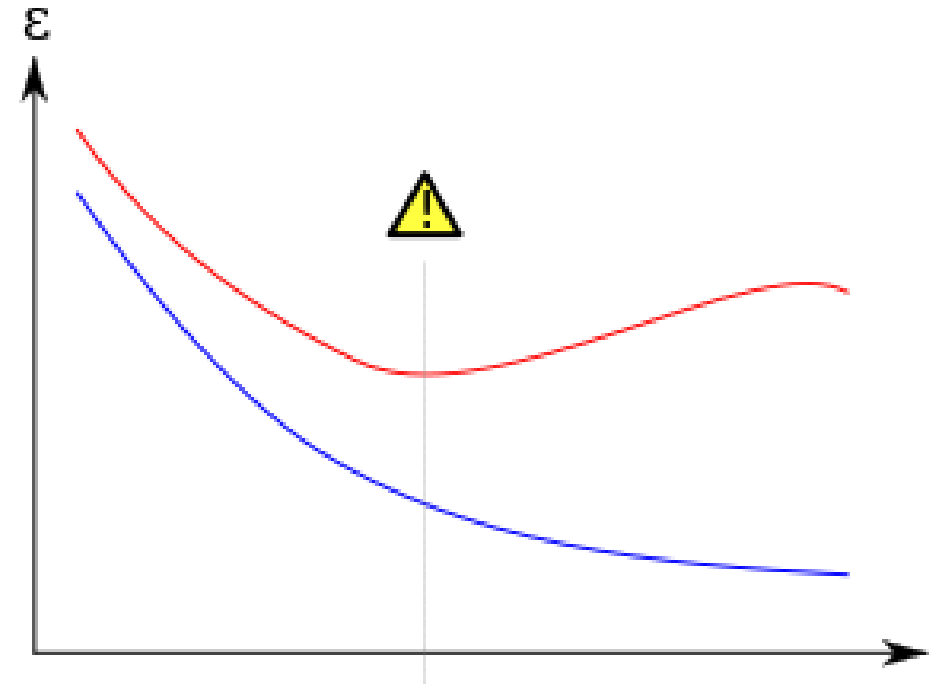
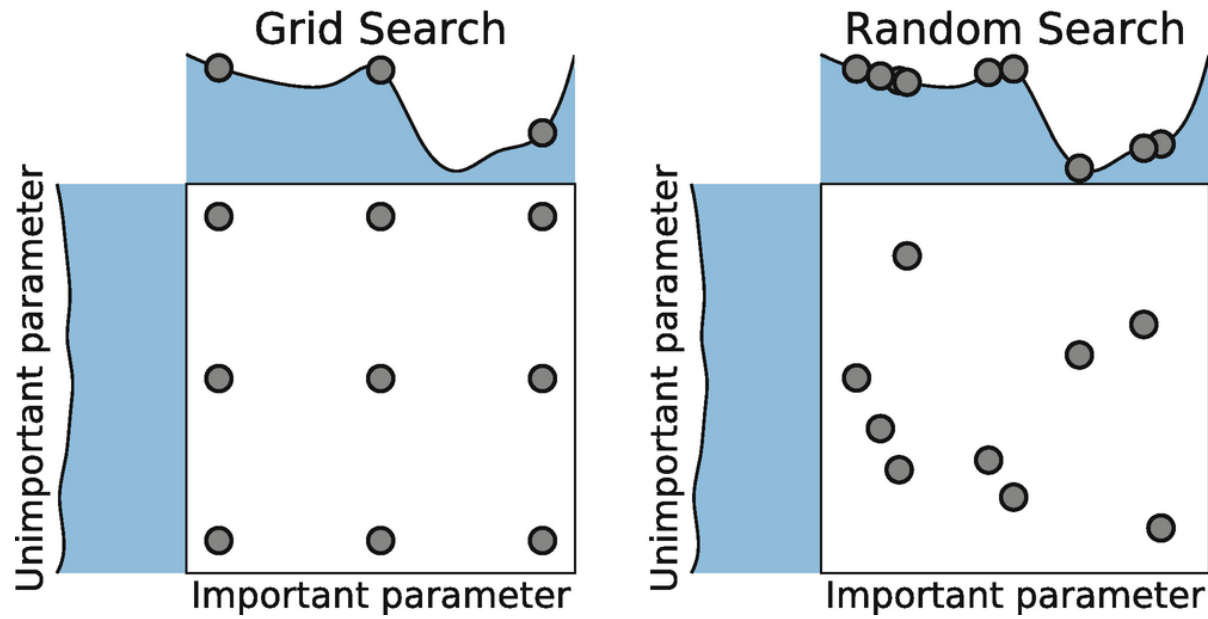
There are many ways to p-hack

1. Stop collecting data once $p < .05$
2. Analyze many measures, but report only those with $p < .05$.
3. Collect and analyze many conditions, but only report those with $p < .05$.
4. Use covariates to get $p < .05$.
5. Exclude participants to get $p < .05$.
6. Transform the data to get $p < .05$.

We prevent p-hacking by using distinct test/train/val datasets



Validation datasets allow us to tune the model hyperparameters



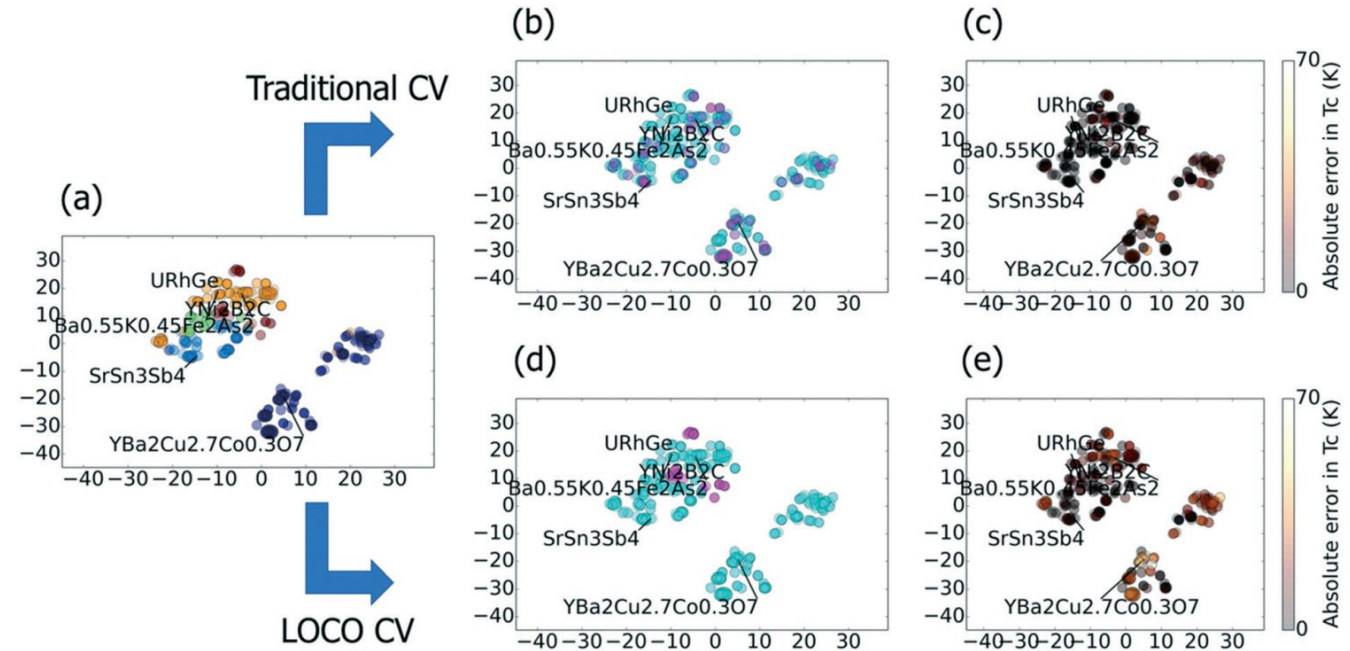
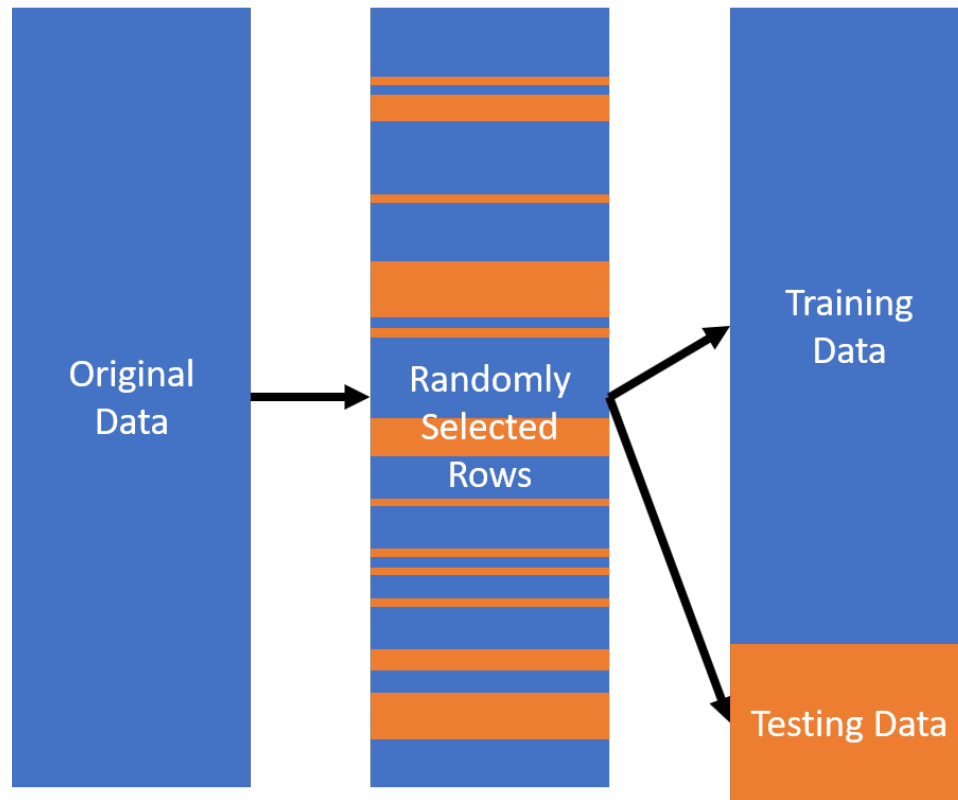
Are there right and wrong ways to split data??

Random

vs

Intentional

Splitting Data for Machine Learning



materials informatics

best practices

